Claim-Eligibility Auditing for Post-hoc Explanations: Synthetic Calibration and Cross-Domain Cases

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Abstract—**Post-hoc explanations can appear coherent even when the fitted model lacks locked-period evidence for the outcome claim being made. We propose a reproducible claim-eligibility audit that separates overall predictive adequacy (Module A), incremental feature-group value (Module B), event-recovery adequacy (Module E), and descriptive diagnostics (Module D). In synthetic stress tests, ungated explanations falsely permit 240/240 structurally invalid runs, whereas the observable A+B+E decision rule reduces false permission to 171/240 while retaining 88.9% sensitivity. The remaining 71.2% false-permission rate and 28.7% specificity show a principled boundary: predictive modules cannot reliably detect leakage, omitted confounding, or distributional violations that preserve or improve predictive performance. We apply the audit to a leakage-safe U.S. crop–state–year panel. A validation-selected Weather-only ExtraTrees model does not pass locked overall adequacy, Full-minus-Metadata incremental weather value also does not pass, and the complete below-trend event-recovery rule does not pass. Consequently, post-hoc explanations are retained only as fitted-function diagnostics. County-level agriculture and PJM demand cases illustrate abstention and permission outcomes without changing the crop decision. The contribution is a locked, same-row, paired-uncertainty protocol that reduces unsupported interpretation while making explicit which validity checks must remain at the study-design stage.**

Index Terms—**claim eligibility, explainable machine learning, temporal validation, crop yield, reproducibility**

I. Introduction

Machine-learning yield forecasts are increasingly built from weather, remote sensing, management, and crop-model variables. Studies report useful predictive systems across large and small panels, diverse model families, and hybrid formulations [1]–[4]. Yet a plausible feature attribution is not causal evidence, and agreement between explanation methods cannot compensate for a model that does not predict the future target reliably. This distinction is especially important for below-trend yield events, where a compelling explanation of an individual fitted prediction can be mistaken for an explanation of the observed loss.

Weather extremes and climate variability are materially associated with crop-yield variation at several spatial scales [5]–[9]. Those findings motivate careful predictive evaluation; they do not license an event-level attribution from this restricted panel. We ask a narrower question: does a leakage-safe residual model have enough future-period predictive adequacy and stability to permit substantive weather-feature-reliance interpretation? The answer is evaluated before XAI is interpreted.

The paper makes four contributions. First, it formulates a study-level claim-eligibility protocol for post-hoc explanations of predictive models. Second, it separates overall predictive adequacy (Module A), the conditional incremental value of weather beyond metadata (Module B), and below-trend event-recovery adequacy (Module E); Weather-only versus Metadata-only remains a representation diagnostic (Module D). Third, it evaluates the rule in simulations with known mechanisms and pre-specified evaluation labels, revealing both abstention behavior and substantial false permission under structural invalidity, plus empirical abstention and permission cases. Fourth, it defines an audit trail that binds each module decision, interval, prediction vector, and target definition to a reproducibility record. Train-only preprocessing and locked testing are protocol safeguards, not claimed as standalone novelty.

II. Related Work

A. Yield Modeling and Detrending

Detrending separates a local historical yield trajectory from residual variation, but the result can depend on the detrending procedure and information available at the evaluation date [10], [11]. We therefore fit each crop–state trend and residual scale on training observations only. Ridge regression, Random Forest, and ExtraTrees provide deliberately distinct linear and tree-based model families [12]–[14]; the implementations use scikit-learn [15]. This is a finite pre-specified comparison, not a search over the locked test period.

B. Explanation Methods and Their Boundary

SHAP and LIME explain a model output under their respective approximation and reference choices [16], [17]. Permutation-style analyses, model reliance, accumulated local effects, and conditional importance likewise describe feature dependence or behavior of a fitted predictor [18]–[20]. They can be useful diagnostic artifacts. However, none turns insufficient prospective predictive adequacy into evidence about an observed yield anomaly. Prior work also shows that post-hoc explanations can be sensitive to perturbation behavior, model randomization, and deletion/retraining benchmarks, and that feature relevance has a causal interpretation problem unless the target estimand and reference distribution are explicit [21]–[25]. Recent surveys, removal-based formulations, counterfactual critiques, and explanation benchmark suites further emphasize that faithfulness and robustness must be evaluated against explicit tasks rather than assumed from visual plausibility [26]–[30]. Those tests ask whether explanations are faithful to model behavior; the audit below asks whether model behavior is eligible to support a locked-period outcome claim.

C. Selective Prediction and Abstention

Selective classification and reject-option systems trade coverage for risk: a predictor may abstain when its risk target is not satisfied [31], [32]. Our unit is different. The decision is not a per-row class label but a study-level permission to interpret a fitted feature family at a specified claim level. The protocol therefore couples a predictive-adequacy module with a paired feature-group value module and reports false permission and false abstention on synthetic scenarios with known data-generating mechanisms and pre-specified evaluation labels.

D. Evaluation and Reproducibility

Random splits can be inappropriate when temporal or hierarchical structure is present [33]. Leakage can create seemingly strong scientific machine-learning results while invalidating the intended forecast claim [34]. Reproducible ML also requires explicit code, data paths, settings, and numerical evidence [35]. Accordingly, every manuscript number is generated from an auditable run: configuration, prediction vectors, bootstrap draws, citation check, and final PDF are versioned outputs.

III. Data and Protocol

A. Panel, Features, and Target

The unit is a crop–state–year row. The processed panel contains 1,257 observations from 1990–2025 for Barley, Canola, Oats, and Wheat across 12 U.S. states: Colorado, Illinois, Iowa, Kansas, Minnesota, Montana, Nebraska, North Dakota, Oklahoma, South Dakota, Texas, and Washington. The observed panel has 40 crop–state series, with available-year counts ranging from 10 to 36 (median 36). Yield in t ha^{-1} comes from USDA NASS Quick Stats [36]; daily maximum/minimum temperature, precipitation, and shortwave radiation come from NASA POWER [37]. The primary design has 35 full-season weather features. A separate sensitivity design adds early-, middle-, and late-season proxies without redefining the target or locked population. Because the primary features require a completed crop-season weather window, the task is a post-season scientific audit of residual-model reliance, not a pre-harvest forecast. Each weather feature is defined by model name, NASA POWER field, aggregation, window, unit, spatial rule, missing handling, and availability in the reproducibility record, which will be released upon acceptance.

For a test interval, each crop–state linear trend and residual scale is estimated using only its associated training rows. A row is eligible only with at least three training observations. The validation period begins with 144 candidate rows and retains 140 after 4 pre-specified exclusions. The locked test starts with 343 candidates, excludes 10, and evaluates 333 rows. Future yields cannot change a previously constructed evaluation target; this property has a unit test and row-level audit. The learned target is the raw train-only residual in t ha^{-1}: $\hat{y}_{c,s,t} = a^{(\text{train})}_{c,s} + b^{(\text{train})}_{c,s} t$, $r_{c,s,t} = y_{c,s,t} - \hat{y}_{c,s,t}$, and $z_{c,s,t} = r_{c,s,t} / \max(\hat{\sigma}_{c,s,\text{train}}, \epsilon)$. The standardized residual defines events, $\mathbb{1}[z_{c,s,t} < \tau]$, only. This distinction prevents a pooled raw-error score from being misread as a standardized event-recovery metric. Min-history rules of 3, 5, 8, and 10 years are reported as sensitivity analyses because both eligibility and the fitted residual scale can change with available history. Three observations are the minimum that leaves one positive residual degree of freedom for estimating a training residual scale; this is not claimed to provide a stable series-specific variance estimate. The model matrices exclude year, yield, trend, residual, standardized residual, event labels, predictions, history length, and residual scale; the no-shortcut schema and overlap audit are retained in the reproducibility record.

B. Selection and Baselines

Ridge $\alpha \in \{1, 10\}$ and Random Forest/ExtraTrees with 160 trees and $\text{min\_samples\_leaf} \in \{1, 2\}$ are evaluated on metadata-only, weather-only, and full representations. Stochastic fits use fixed seeds $\{7, 17, 27, 37, 47\}$; numeric imputation/scaling, categorical encoding, and model fitting occur within the training fold. Mean per-row seed-aggregated validation RMSE chooses the configuration, with lexicographic tie-breaking and no locked-test access. Seed-level RMSE standard deviations are reported so tiny differences are not presented as deterministic wins.

The final audit compares learned predictions on identical rows to zero residual, training-mean residual, and crop training-mean residual baselines. Prediction vectors are hashed and pairwise maximum absolute differences are retained, so any nearly coincident baselines are demonstrated rather than assumed. We use 2,000 deterministic year-block bootstrap replicates and percentile 95% intervals over the ten locked year-block units. With only ten locked year-block units, interval precision and statistical power are necessarily limited. We nevertheless retain year-block paired resampling because it preserves the cross-sectional dependence shared by crop–state rows within the same calendar year. All 2,000 bootstrap replicates were valid and retained. We do not implement the stationary bootstrap itself; we cite it for the dependent-resampling motivation and resample complete calendar years as paired blocks [38]. State-year and crop–state cluster schemes are reported as pre-specified sensitivity analyses.

C. Pre-specified Claim Modules

Let P^* denote the locked target population implied by the split, target construction, feature set, and evaluation period. Module A estimates $\theta_A = \text{RMSE}_{P^*}(f^*) - \text{RMSE}_{P^*}(f_0)$, where f^* is the validation-selected model and f_0 is the zero-residual baseline. Module B estimates $\theta_B = \text{RMSE}_{P^*}(f_{M+W}) - \text{RMSE}_{P^*}(f_M)$, the incremental weather value beyond metadata. The reported estimates replace P^* by

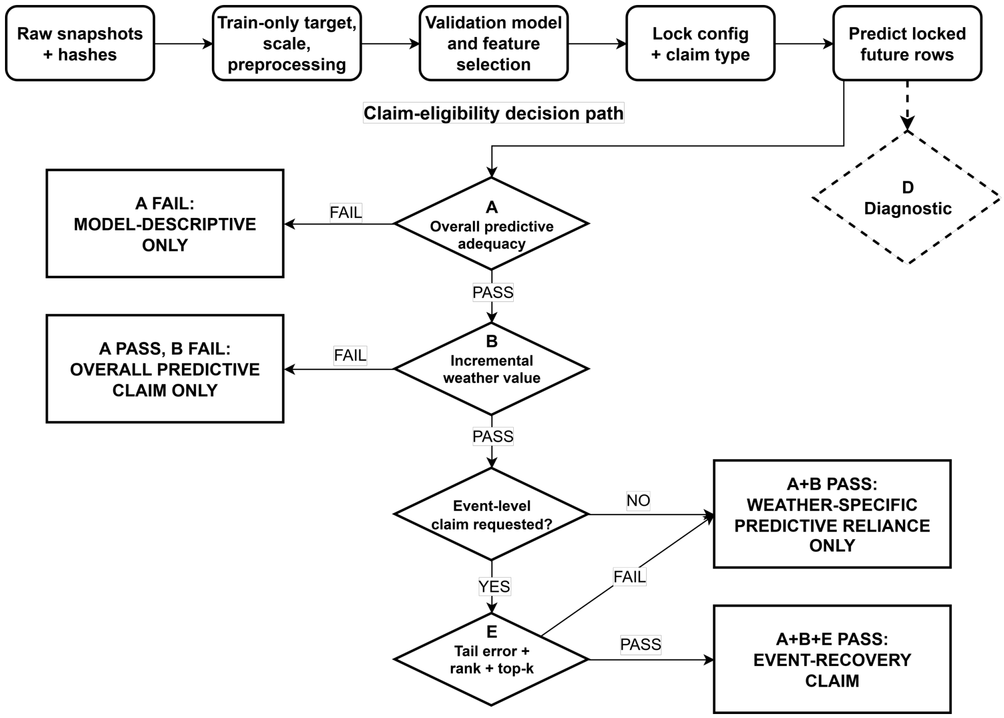

Fig. 1. Sequential claim-eligibility workflow. Module A determines whether claims may extend beyond model description, Module B controls weather-specific predictive reliance, and Module E is evaluated only for an event-level claim. Module E not passing blocks event-recovery interpretation but retains the weather-specific predictive-reliance claim when Modules A and B pass. Module D is a diagnostic outside the permission path.

the locked empirical rows, use paired year-block resampling, and pass only when the upper 95% interval is below zero.

Module E is invoked only for below-trend event claims. Its primary tail is $z < -1$; $z < -1.5$ and $z < -2$ are sensitivities because their smaller samples reduce power. The event module requires tail RMSE/MAE improvement, rank recovery with a positive lower bootstrap bound and within-year permutation $p < .05$, and top-k recovery above chance by lift, hypergeometric probability, permutation p, and year-block lift interval. Module D is the Weather-only versus Metadata-only representation diagnostic and is recorded outside the permission path. Module D is retained as a general representation diagnostic for settings in which metadata carries predictive information; it coincides numerically with Module A only in this panel because Metadata-only predictions collapse to the zero-residual baseline. The fixed claim-eligibility audit is summarized in Table I.

IV. Results

A. Validation and Locked Test

The selected validation configuration is ExtraTrees with Weather-only features. The complete grid, seed-level metrics, tie rule, and locked-test access flag are retained in the selection record. Module A uses this single overall validation-selected configuration. Module B is a pre-specified fixed-architecture contrast: Full and Metadata-only predictions reuse the Module A configuration ID and swap only the feature family. Their metrics and row/target/prediction hashes are recorded in the selection manifest; they are not candidates for replacing Module A after the locked period is observed.

The selected weather-only ExtraTrees configuration ranks first in 3/5 stochastic validation seeds. Its advantage is small relative to several seed standard deviations, so we retain the pre-specified selection without claiming clear superiority. The one-standard-error flag and every seed-level score are retained; no final-test statistic changes this selection.

On the locked 2016–2025 test, the selected model has $R^2 = -0.014$ and RMSE 0.669 t ha^{-1}. Its paired RMSE difference from zero crosses zero, so Module A does not pass. The selected configuration deteriorates from validation RMSE 0.384 in 2012–2015 to locked-test RMSE 0.669 in 2016–2025, consistent with the temporal instability summarized below. Table II includes all same-task naive baselines and the selected configuration across feature families. Module B compares Full versus Metadata-only on the same locked rows. Module D compares Weather-only and Metadata-only as an exploratory representation diagnostic outside the permission path. All comparisons use paired predictions on identical locked rows. Modules A and B do not pass; Module D shows no clear difference and remains diagnostic.

The Module B interval is computed directly from paired Full-minus-Metadata predictions on the same locked rows. It is numerically identical to Full-minus-zero in this panel because the Metadata-only prediction vector coincides with the zero-residual baseline to floating-point precision. The Metadata-only vector nearly coincides with the zero-residual baselines

TABLE I

FIXED CLAIM-ELIGIBILITY MODULES AND LOCKED CROP-PANEL VERDICTS. MODULE D (WEATHER-ONLY VERSUS METADATA-ONLY) IS A DESCRIPTIVE REPRESENTATION DIAGNOSTIC OUTSIDE THE PERMISSION PATH AND IS THEREFORE NOT A REQUIRED ROW. FULL CONFIGURATION DETAILS ARE IN THE REPRODUCIBILITY MANIFEST.

Mod.	Question/contrast	Pass rule	Current estimate	V5 verdict	Claim consequence
A	Selected Weather-only vs zero	upper paired CI < 0	-0.005 [-0.019, 0.009]	DOES NOT PASS	model description only
B	Full vs Metadata-only	upper paired CI < 0	-0.012 [-0.029, 0.002]	DOES NOT PASS	no weather-specific reliance claim
E	Primary tail error + rank + top-k	all checks pass	RMSE -0.043 [-0.074, 0.007]; rank 0.180; top-10 1/10	DOES NOT PASS	no event-recovery claim

on the locked rows. The target is an OLS residual constructed separately within each crop–state series, so its training mean is approximately zero within each fitted series. Metadata features are time-invariant context variables (latitude, longitude, region, and crop) and provide no within-series temporal signal. In the locked ExtraTrees configuration, the maximum absolute prediction difference is 1.32×10^{-13} versus the zero-residual vector, 1.13×10^{-13} versus the training-mean baseline, and 1.05×10^{-13} versus the crop-training-mean baseline. Consequently, Module D is numerically identical to Module A in this panel, although the contrasts answer different conceptual questions. Module B remains distinct because the Full model can use weather jointly with metadata.

B. Event-Detection Diagnostics

Event detection is evaluated over all 333 locked rows, not only a tail preselected by the observed outcome. The score is negative predicted residual and the score threshold is fixed at zero before final evaluation. Event scores retain weak global discrimination: ROC-AUC is 0.641 for the primary $z < -1$ tail and 0.672 for the two severe-tail sensitivities; primary-tail PR-AUC is 0.338 versus prevalence 0.219. This does not contradict the module decision. ROC-AUC summarizes pairwise ordering over all locked rows, whereas Module E asks whether the model improves paired error and recovers the pre-specified tail beyond rank and top-k null checks. The available discrimination is diagnostically positive but insufficient for the requested observed-event claim.

C. Below-Trend Event Module and Robustness

The broad below-trend subset contains 73 locked-test rows. The audit artifact separates error, rank-null, and chance-adjusted event-recovery components. The primary $z < -1$ top-10 lift is below random expectation and its permutation result is not significant; rank uncertainty also crosses zero. Module E therefore does not pass without relying on a directional-sign statistic. Because Modules A and B do not pass, neither the local nor global explanation outputs can be used to support observed-event or weather-specific predictive claims in the primary analysis. The components of Module E target distinct evidentiary requirements: tail RMSE/MAE test error magnitude, rank statistics test ordering, and chance-adjusted top-k checks test prioritization of the most important events. The module is not mechanically impossible to pass: in the synthetic benchmark, moderate-signal, strong-signal, and valid imbalanced-tail regimes pass Module E in 30/30 seeds.

The locked Weather-only model still produces coherent fitted-function descriptions: grouped SHAP mass is concentrated in weather features, and the local decomposition in Fig. 2 assigns one below-trend row to recognizable weather groups. These outputs describe how the fitted predictor varies with its inputs. They are therefore reported as model diagnostics rather than evidence about observed yield losses.

Audit-record sensitivity checks for temporal stage features, crop, spatial, and target protocol mark stability boundaries rather than reversing the central decision.

D. Sensitivity and Traceability Summary

Sensitivity checks show where future designs may improve, not reselect after the locked period. The audit record retains History 3, History 5, History 8, History 10, Standardized target, Huber detrending, HistGradientBoosting, ElasticNet, cluster-resampling, prefix learning, retrospective-target, and state-heterogeneity diagnostics for release. Together these records cover 11 pre-specified sensitivity categories and 51 recorded rows. None changes the primary Module A/B/E decision or converts a descriptive diagnostic into event-recovery evidence. Figure 4 shows descriptive state-level heterogeneity; these subgroup contrasts do not replace the panel-level Module A decision.

TABLE II

LOCKED 2016–2025 SAME-TASK AUDIT. LEARNED-MODEL PREDICTIONS ARE SEED AGGREGATED. ΔRMSE IS MODEL MINUS ZERO-RESIDUAL RMSE WITH PAIRED 95% YEAR-BLOCK INTERVALS; RMSE IS T HA^{-1}. THE MODULE B CONTRAST IS COMPUTED DIRECTLY AS FULL-MINUS-METADATA; IN THIS PANEL IT MATCHES FULL-MINUS-ZERO BECAUSE METADATA-ONLY PREDICTIONS COINCIDE WITH THE ZERO-RESIDUAL VECTOR TO FLOATING-POINT PRECISION.

Model	Features	n	Seeds	R^2	RMSE	ΔRMSE vs zero [95% CI]
Crop train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Train mean	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Zero residual	Baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
ExtraTrees	Full	333	5	0.006	0.662	-0.012 [-0.029, 0.002]
ExtraTrees	Metadata-only	333	5	-0.031	0.674	0.000 [0.000, 0.000]
ExtraTrees	Weather-only	333	5	-0.014	0.669	-0.005 [-0.019, 0.009]

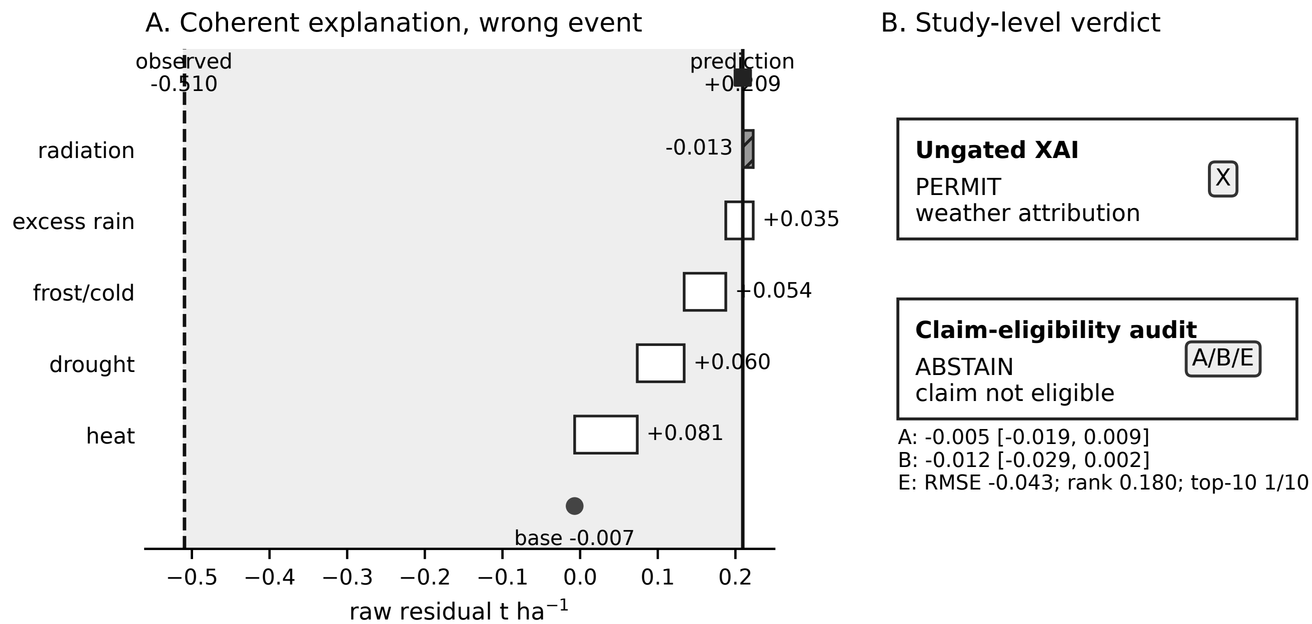

Fig. 2. Corrected agricultural explanation diagnostic for the locked Weather-only Barley–Colorado 2016 row. The local grouped SHAP decomposition is shown on the same residual scale as the observed outcome: base value approximately -0.008, prediction $+0.209$, and observed residual -0.510 t ha^{-1}. The verdict panel reports the claim-eligibility consequence: because Modules A, B, and E do not pass on locked outcomes, the explanation is a fitted-function diagnostic rather than evidence about the observed loss.

V. CALIBRATION AND EXTERNAL CASES

A. Synthetic Benchmark

The claim-eligibility audit is also evaluated where the mechanism and evaluation label are known before scoring. A repeated synthetic temporal benchmark covers 14 regimes over 30 seeds each, including correlated features, omitted confounding, temporal and geographic shift, and leakage. GT labels define six permissible regimes and are used only for evaluation; audit permission is computed from observable Modules A, B, and E. Ungated explanations falsely permit 240 of 240 invalid runs (100.0%, 95% CI [98.4,100.0]); the observable A+B+E rule falsely permits 171/240 invalid runs (71.2%, 95% CI [65.2,76.6]) while falsely abstaining on 20/180 valid runs (11.1%, 95% CI [7.3,16.5]). Its permission rate is 78.8%, sensitivity 88.9%, and specificity 28.7%.

The observable modules cannot reliably identify all structurally invalid regimes: omitted confounding, geographic shift, leakage, and correlated features are permitted in 30/30 seeds, and temporal drift in 29/30. These are false permissions, not successful abstentions.

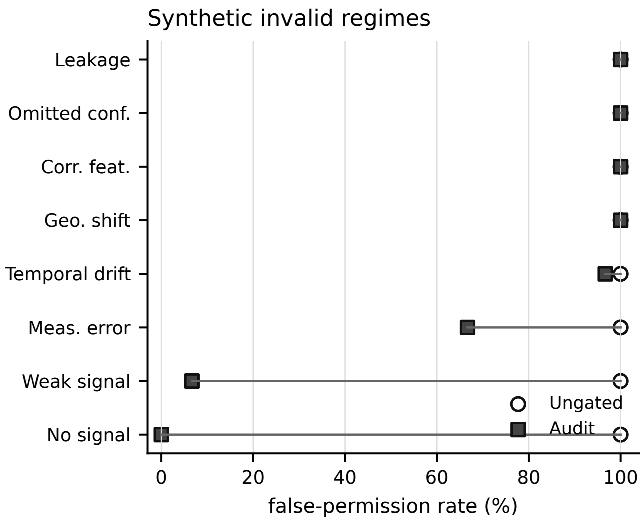

Fig. 3. Synthetic calibration of ungated explanation permission versus the observable A+B+E audit rule across structurally invalid regimes. GT labels are used only for evaluation, not as inputs to the audit decision.

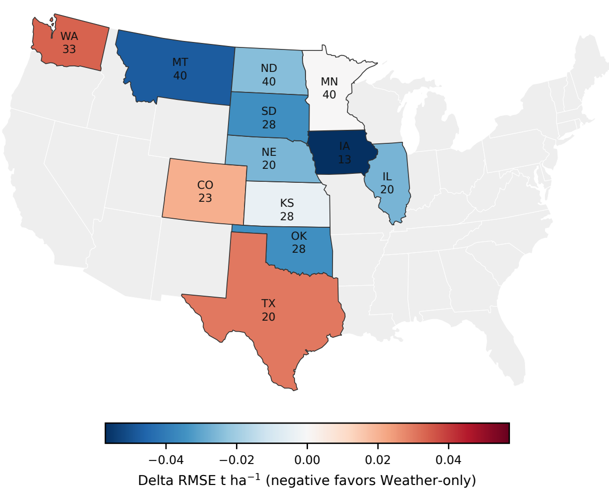

Fig. 4. State-level heterogeneity on the locked 2016–2025 crop panel. Colors show Weather-only minus zero-residual ΔRMSE; negative values favor Weather-only. Labels report state abbreviation and locked-row count. Gray states are outside the study panel. Subgroup estimates are descriptive and not separate module decisions.

B. Agricultural External-Resolution Check

We also implemented a separate county-level winter-wheat check rather than assuming that a finer panel would rescue the state-level result. The locked pre-model population contains 574 counties with at least ten training-period observations from 2000–2025, official county yields, static county interior points, and annual NASA POWER weather aggregates. Validation selected a weather-only ExtraTrees residual model from Ridge and ExtraTrees metadata, weather, and full-feature candidates. On the untouched 2022–2025 holdout, its RMSE was 13.47 versus 13.78 bushels per acre for the zero-residual baseline, but the paired year-block 95% CI for the difference was $[-0.67, 0.26]$. Module B, which compares Full with Metadata-only, passes (ΔRMSE $= -0.71$, 95% CI $[-1.23, -0.34]$), but Module A remains inconclusive. The county analysis therefore reaches a model-descriptive-only outcome and serves as a robustness case rather than evidence of predictive success or cross-resolution transfer.

C. Cross-Domain Sanity-Check Permission Case

The PJM analysis is a cross-domain sanity-check permission case for the core two-stage audit rather than a transfer claim. Its target is continuous electricity demand, so the crop-specific extreme-event module is not invoked. Public daily PJM electricity-demand and day-ahead forecast records from EIA Form EIA-930 [39] are fit with Calendar-only ExtraTrees features on January–September 2024 and evaluated on locked October–December records; Full adds the day-ahead demand forecast. Module A compares Full with a pre-specified train-period mean-demand naive baseline and passes: locked RMSE is 77,521 versus 335,037 MWh, with paired bootstrap ΔRMSE 95% CI $[-296.7, -221.8] \times 10^3\,\mathrm{MWh}$. Module B compares Full with Calendar-only and also passes: Calendar-only RMSE is 227,417 MWh and the Full-minus-Calendar paired 95% CI is $[-186.0, -119.3] \times 10^3\,\mathrm{MWh}$. Because both pass, permutation importance is reported for locked PJM rows. The XAI manifest binds row IDs, hashes, seed handling, and output scale; agricultural XAI outputs remain descriptive because the required agricultural modules do not pass. These are predictive-reliance interpretations, not causal or transfer claims. This intentionally easy positive case checks that the audit is not always-abstaining; synthetic regimes define the decision-rule boundary.

VI. Discussion

A. Claim Scope and Multiplicity

The synthetic benchmark clarifies the scope of the proposed audit. Relative to ungated explanation, the observable A+B+E rule reduces false permission from 240/240 to 171/240 invalid runs. The remaining false permissions are concentrated in regimes such as leakage, omitted confounding, correlated features, and geographic or temporal shift, which may preserve or even improve predictive performance. Predictive auditing should therefore be interpreted as a necessary claim-discipline layer, not as a detector of structural validity; design-stage leakage, confounding, measurement, and distribution-shift checks remain independently required.

Several analyses are locally favorable. The eight- and ten-year history filters improve paired error on a smaller 313-row population; the 2013–2015 rolling fold passes, whereas the 2007–2009 fold significantly favors the baseline; severe tails improve RMSE and MAE but do not pass rank and top-k recovery; and the county analysis passes Module B while Module A remains inconclusive. These results identify design conditions under which predictive signal may be stronger, but they either change the population, represent non-primary temporal folds, or do not pass another required module. We therefore use them to motivate future study design rather than to replace the primary locked test.

The audit separates primary decisions from descriptive measurements before the locked period is examined. Module A, primary Module B, and the primary $z < -1$ Module E define the below-trend event permission rule; more severe tails, alternative histories, detrending choices, resampling units, stage proxies, crop/state diagnostics, and expanded model families are sensitivity or diagnostic evidence. They clarify robustness and design trade-offs but are not extra chances to declare the primary claim successful.

B. Reproducibility and Traceability

The reproducibility record contains input hashes, target construction, feature availability, forbidden-column checks, locked configuration, seeds, row IDs, predictions, bootstrap draws, and table inputs, so paired comparisons can be recomputed on identical locked rows.

C. Limitations and External Validity

State-representative weather can miss within-state exposure, weather covariates are correlated, and the panel lacks soil, irrigation, cultivar, management, and phenology data. The primary interval resamples only ten locked year-block units, and cluster sensitivities encode different dependence assumptions without removing that limitation. This negative result limits the present target, resolution, features, and model family; it does not establish that weather lacks scientific relevance for yield or that a stronger model cannot exist. The corrected synthetic benchmark also shows that predictive adequacy and feature-group value cannot identify omitted confounding, leakage, correlated-feature structures, or unobserved distributional violations by themselves. GT labels are used only for evaluation, not for audit decisions. The audit asks whether the validation-selected fitted model has sufficient locked evidence to support the requested claim. The separately locked county check shows that finer resolution does not automatically rescue overall adequacy, while the PJM permission case demonstrates that the audit does not always abstain when signal is strong. Future work needs prospectively versioned inputs, finer exposure measurements, design-stage validity checks, and an independent future test interval.

VII. Conclusion

The agricultural audit supports model description only: Modules A, B, and E do not pass on the locked state panel, and the separately locked county check remains inconclusive under Module A. More generally, the audit reduces unsupported permission relative to ungated explanation but does not establish structural validity. The synthetic reduction from 240/240 to 171/240 false permissions demonstrates practical claim discipline, while the remaining false permissions show why leakage, omitted confounding, measurement validity, and distribution-shift checks must be enforced at the design stage.

References

[1] D. Paudel, H. Boogaard, A. de Wit, S. Janssen, S. Osinga, C. Pylianidis, and I. N. Athanasiadis, "Machine learning for large-scale crop yield forecasting," *Agricultural Systems*, vol. 187, p. 103016, 2021.

[2] M. Meroni, F. Waldner, L. Seguini, H. Kerdiles, and F. Rembold, "Yield forecasting with machine learning and small data: What gains for grains?" *Agricultural and Forest Meteorology*, vol. 308–309, p. 108555, 2021.

[3] G. Leng and J. W. Hall, "Predicting spatial and temporal variability in crop yields: an inter-comparison of machine learning, regression and process-based models," *Environmental Research Letters*, vol. 15, p. 044027, 2020.

[4] T. van Klompenburg, A. Kassahun, and C. Catal, "Crop yield prediction using machine learning: A systematic literature review," *Computers and Electronics in Agriculture*, vol. 177, p. 105709, 2020.

[5] C. Lesk, P. Rowhani, and N. Ramankutty, "Influence of extreme weather disasters on global crop production," *Nature*, vol. 529, pp. 84–87, 2016.

[6] D. K. Ray, J. S. Gerber, G. K. MacDonald, and P. C. West, "Climate variation explains a third of global crop yield variability," *Nature Communications*, vol. 6, p. 5989, 2015.

[7] W. Schlenker and M. J. Roberts, "Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change," *Proceedings of the National Academy of Sciences*, vol. 106, pp. 15 594–15 598, 2009.

[8] M. Zampieri, A. Ceglar, F. Dentener, and A. Toreti, "Wheat yield loss attributable to heat waves, drought and water excess," *Environmental Research Letters*, vol. 12, p. 064011, 2017.

[9] F. Schierhorn, M. Hofmann, T. Gagalyuk, I. Ostapchuk, and D. Müller, "Machine learning reveals complex effects of climatic means and weather extremes on wheat yields during different plant developmental stages," *Climatic Change*, vol. 169, p. 39, 2021.

[10] J. Lu, G. J. Carbone, and P. Gao, "Detrending crop yield data for spatial visualization of drought impacts in the United States, 1895–2014," *Agricultural and Forest Meteorology*, vol. 237–238, pp. 196–208, 2017.

[11] H. Meng and L. Qian, "Performances of different yield-detrending methods in assessing the impacts of agricultural drought and flooding: A case study in the middle-and-lower reach of the Yangtze River, China," *Agricultural Water Management*, vol. 296, p. 108812, 2024.

[12] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, pp. 55–67, 1970.

[13] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.

[14] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Machine Learning*, vol. 63, pp. 3–42, 2006.

[15] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[16] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[17] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why Should I Trust You? Explaining the Predictions of Any Classifier," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.

[18] A. Fisher, C. Rudin, and F. Dominici, "All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously," *Journal of Machine Learning Research*, vol. 20, no. 177, pp. 1–81, 2019. [Online]. Available: https://jmlr.org/papers/v20/18-760.html

[19] D. W. Apley and J. Zhu, "Visualizing the effects of predictor variables in black box supervised learning models," *Journal of the Royal Statistical Society: Series B*, vol. 82, pp. 1059–1086, 2020.

[20] C. Strobl, A.-L. Boulesteix, A. Zeileis, and T. Hothorn, "Bias in random forest variable importance measures: Illustrations, sources and a solution," *BMC Bioinformatics*, vol. 9, p. 307, 2008.

[21] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems*, vol. 31, 2018. [Online]. Available: https://papers.nips.cc/paper/8160-sanity-checks-for-saliency-maps

[22] S. Hooker, D. Erhan, P.-J. Kindermans, and B. Kim, "A benchmark for interpretability methods in deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: https://proceedings.neurips.cc/paper/2019/hash/fe4b8556000d0f0cae99daa5c5c5a410-Abstract.html

[23] C.-K. Yeh, C.-Y. Hsieh, A. S. Suggala, D. I. Inouye, and P. K. Ravikumar, "On the (in)fidelity and sensitivity of explanations," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: https://papers.neurips.cc/paper/9278-on-the-infidelity-and-sensitivity-of-explanations

[24] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods," in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 2020, pp. 180–186.

[25] D. Janzing, L. Minorics, and P. Blöbaum, "Feature relevance quantification in explainable AI: A causal problem," in *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 108. PMLR, 2020, pp. 2907–2916. [Online]. Available: https://proceedings.mlr.press/v108/janzing20a.html

[26] I. Covert, S. Lundberg, and S.-I. Lee, "Explaining by removing: A unified framework for model explanation," *Journal of Machine Learning Research*, vol. 22, no. 209, pp. 1–90, 2021. [Online]. Available: https://jmlr.org/papers/v22/20-1316.html

[27] M. T. Keane, E. M. Kenny, E. Delaney, and B. Smyth, "If only we had better counterfactual explanations: Five key deficits to rectify in the evaluation of counterfactual XAI techniques," in *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence*, 2021, pp. 4466–4474.

[28] C. Agarwal, S. Krishna, E. Saxena, M. Pawelczyk, N. Johnson, I. Puri, M. Zitnik, and H. Lakkaraju, "OpenXAI: Towards a transparent evaluation of model explanations," in *Thirty-sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2022. [Online]. Available: https://openreview.net/forum?id=MU2495w47rz

[29] M. Nauta, J. Trienes, S. Pathak, E. Nguyen, M. Peters, Y. Schmitt, J. Schlötterer, M. van Keulen, and C. Seifert, "From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable AI," *ACM Computing Surveys*, vol. 55, no. 13s, pp. 295:1–295:42, 2023.

[30] A. Hedström, L. Weber, D. Krakowczyk, D. Bareeva, F. Motzkus, W. Samek, S. Lapuschkin, and M. M.-C. Höhne, "Quantus: An explainable AI toolkit for responsible evaluation of neural network explanations and beyond," *Journal of Machine Learning Research*, vol. 24, no. 34, pp. 1–11, 2023. [Online]. Available: https://jmlr.org/papers/v24/22-0142.html

[31] R. El-Yaniv and Y. Wiener, "On the foundations of noise-free selective classification," *Journal of Machine Learning Research*, vol. 11, no. 53, pp. 1605–1641, 2010. [Online]. Available: https://jmlr.org/papers/v11/el-yaniv10a.html

[32] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://papers.neurips.cc/paper/7073-selective-classification-for-deep-neural-networks

[33] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schroder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, pp. 913–929, 2017.

[34] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, p. 100804, 2023.

[35] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Lariviere, A. Beygelzimer, F. d'Alche Buc, E. Fox, and H. Larochelle, "Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021. [Online]. Available: https://jmlr.org/papers/v22/20-303.html

[36] USDA National Agricultural Statistics Service, "Quick stats database," 2026, accessed 2026-07-18. [Online]. Available: https://quickstats.nass.usda.gov/

[37] NASA Prediction Of Worldwide Energy Resources Project, "POWER data access viewer and API," 2026, accessed 2026-07-18. [Online]. Available: https://power.larc.nasa.gov/

[38] D. N. Politis and J. P. Romano, "The stationary bootstrap," *Journal of the American Statistical Association*, vol. 89, no. 428, pp. 1303–1313, 1994.

[39] U.S. Energy Information Administration, "EIA-930, hourly electric grid monitor," 2026, accessed 2026-07-19. [Online]. Available: https://www.eia.gov/electricity/gridmonitor/about